

Michael Miller Yoder

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<https://michaeltilleryoder.github.io>

EDUCATION	Carnegie Mellon University , Pittsburgh, PA	
	<i>Ph.D.</i> , Language and Information Technologies	2021
	Advisor: Carolyn P. Rosé	
	Thesis: “Computational models of identity presentation in language” PDF	
	M.S. , Language Technologies	2017
	Goshen College , Goshen, IN	
	<i>B.A.</i> , Computer Science	2013
	Minors: English, International Studies	
PROFESSIONAL EXPERIENCE	Teaching Assistant Professor	University of Pittsburgh Pittsburgh, PA
	2024–present	
	Teaching-track faculty member in the School of Computing and Information. Focusing on data science instruction to students from a wide variety of backgrounds in both in-person and online formats.	
	Postdoctoral Research Associate	Carnegie Mellon University Pittsburgh, PA
	2022–2024	
	Researcher with the Center for Applied Research on Targeted Violence (formerly the Collaboratory Against Hate), an interdisciplinary initiative of Carnegie Mellon University and the University of Pittsburgh, as well as the Center for Informed Democracy and Social Cybersecurity at Carnegie Mellon University. Developed new methods and analyses for tracking and detecting hate speech in online contexts, incorporating insights from the social sciences. Advised by Prof. Kathleen M. Carley in the Software and Societal Systems Department and Prof. David West Brown in the Department of English.	
	Part-Time Instructor	University of Pittsburgh Pittsburgh, PA
	2023–2024	
	Instructor for graduate-level NLP course in the Department of Computer Science.	
	Graduate Research Assistant	Carnegie Mellon University Pittsburgh, PA
	2015–2021	
	Applied computational models of discourse to study social meaning in language, largely using social media datasets. Advised by Prof. Carolyn P. Rosé in the Language Technologies Institute.	
	Software Engineering Intern	Yahoo Research/Tumblr New York, NY
	2019	
	Detected communities of Tumblr users from social network structure. Explored community-specific language use. Advised by Dr. Kapil Thadani.	
	Research Programmer/Analyst	Carnegie Mellon University Pittsburgh, PA
	2015	
	Designed and implemented neuroimaging database to support machine learning research. Advised by Prof. Timothy Verstynen in the Department of Psychology.	
	Software Developer & IT Administrator	GSAREH Austin, TX area
	2014	
	Developed applications to recognize and map named entities in English and Arabic articles and social media. Managed software development and IT administration of a small GIS startup named Geospatial Applied Research Experts House.	

TEACHING

Human Language Technologies HTML University of Pittsburgh
Spring 2025, Spring 2026 Pittsburgh, PA
Instructor for undergraduate-level NLP course. Developed lectures, in-class activities and coding notebooks, quizzes, homework assignments, and an exam. Guided open-ended student-led projects.

Applied Predictive Modeling University of Pittsburgh
Spring 2025, Spring 2026 Online
Instructor for graduate-level online course on machine learning methods and applied, practical modeling skills, part of both the Applied Data-Driven Methods graduate certificate for working professionals. Guided course projects in which students analyzed datasets of their own choosing.

Introduction to University of Pittsburgh
Natural Language Processing HTML
Fall 2023, Spring 2024, Fall 2024, Fall 2025 Pittsburgh, PA
Instructor for graduate-level NLP course. Developed coursework, including lectures, quizzes, and homework assignments. Guided hands-on, student-led projects.

Introduction to University of Pittsburgh
Data-Centric Computing
Fall 2024, Fall 2025, Spring 2026 Online
Instructor for graduate-level online course developed as part of the Applied Data-Driven Methods graduate certificate and Master of Data Science program for working professionals. Guided up to 200 students in an online format through an introduction to programming and data analysis with Python all the way to course projects in which students complete a data analysis pipeline on real-world data.

Computational Ethics for Carnegie Mellon University
Natural Language Processing HTML
Spring 2020 Pittsburgh, PA
Teaching assistant for newly developed NLP ethics course with Profs. Yulia Tsvetkov and Alan Black. Gave guest lectures, developed homeworks, and graded coursework.

Applied Machine Learning Carnegie Mellon University
Spring 2019 Pittsburgh, PA
Teaching assistant with Prof. Carolyn P. Rosé for a hands-on machine learning course developed for a broad range of students, including those without computer science backgrounds. Gave guest lectures, advised students in office hours, and graded coursework.

Introduction to Artificial Intelligence Goshen College
Spring 2013 Goshen, IN
Selected as a tutor for introductory artificial intelligence course. Assisted students one-on-one with AI concepts and coursework.

GRANTS

Analyzing Harmful Language and Social Dynamics in Online Gaming Voice Chat
Michael Miller Yoder, Dan Villarreal, Dmitriy Babichenko, Lorraine Li, Scott Kiesling
Pitt Cyber, University of Pittsburgh, 2025
\$4500

Linear Algebra and its Machine Learning Applications
Marina Barsky, Michael Miller Yoder
Nancy Tannery Grant for Open Educational Resources, University of Pittsburgh, 2025
\$5000

	Identifying and Analyzing Influencer Marketing of Pharmaceutical Products on Social Media Ryan Shi, Michael Miller Yoder, Olga Kravchenko Pitt Cyber, University of Pittsburgh, 2024 \$5100
RESEARCH INTERESTS	Natural language processing, computational social science, data science, AI ethics, digital humanities, sociolinguistics, hate speech, language and identity
JOURNAL ARTICLES	FATe of Bots: Ethical Considerations of Social Bot Detection Lynnette Hui Xian Ng, Ethan Pan, Michael Miller Yoder, Kathleen Carley <i>ACM Journal on Responsible Computing</i>, 3(2), Article 9, 2026. PDF Modes of toxic behavior and game design considerations in online multi-player games Lynnette Hui Xian Ng, Adrian Xuan Wei Lim, Michael Miller Yoder <i>First Monday</i>, 30(6), 2025. The “chanification” of white supremacist extremism Michael Miller Yoder, David West Brown, Kathleen M. Carley <i>Computational and Mathematical Organization Theory</i>, 2024. PDF Implicit Antisemitism and COVID-19 Conspiracy Theory on Twitter: Linking Narratives in a Mixed-Methods Study Grace Garner, Madeleine McGrann, Maja Lynn, Daniel Klug, Rachel Kranson, Michael Miller Yoder <i>Jewish Film & New Media: An International Journal</i>, 11(2), 183–213, 2023. HTML An empirical analysis of high school students’ practices of modelling with unstructured data Shiyan Jiang, Amato Nocera, Cansu Tatar, Michael Miller Yoder, Jie Chao, Kenia Wiedemann, William Finzer, Carolyn P. Rosé <i>British Journal of Educational Technology</i>, 1–20, 2022. HTML. Unpacking a political icon: “Bike lanes” and orders of indexicality Michael Miller Yoder, Barbara Johnstone <i>Discourse & Communication</i>, 12(2), 192–208, 2018.
CONFERENCE PUBLICATIONS	Measuring Dimensions of Self-Presentation in Twitter Bios and their Links to Misinformation Sharing Navid Madani, Rabiraj Bandyopadhyay, Michael Miller Yoder, Stephan D. McCabe, Briony Swire-Thompson, Kenneth Joseph <i>Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)</i>. 2025. PDF A Weakly Supervised Classifier and Dataset of White Supremacist Language Michael Miller Yoder, Ahmad Diab, David West Brown, Kathleen M. Carley <i>Proceedings of the Association for Computational Linguistics (ACL)</i>. 2023. PDF How Hate Speech Varies by Target Identity: A Computational Analysis Michael Miller Yoder, Lynnette Hui Xian Ng, David West Brown, Kathleen M. Carley <i>Proceedings of the Conference on Computational Natural Language Learning (CoNLL)</i>, 2022. PDF. Modeling Unstructured Data: Teachers as Learners and Designers of Technology-enhanced Artificial Intelligence Curriculum

Cansu Tatar, Michael Miller Yoder, Madeleine Coven, Kenia Wiedemann, Jie Chao, William Finzer, Shiyan Jiang, Carolyn P. Rosé
Proceedings of the International Society of the Learning Sciences, 2021. PDF.

Phans, Stans and Cishets: Self-presentation Effects on Content Propagation in Tumblr

Michael Miller Yoder, Qinlan Shen, Yansen Wang, Alex Coda, Yunseok Jang, Yale Song, Kapil Thadani, Carolyn P. Rosé
Proceedings of the ACM Conference on Web Science (WebSci), 2020. PDF.

Perceptions of Censorship and Moderation Bias in Political Debate Forums

Qinlan Shen, Michael Miller Yoder, Yohan Jo, Carolyn P. Rosé
Proceedings of the International AAAI Conference on Web and Social Media (ICWSM), 2018. PDF.

Roles and Success in Wikipedia Talk Pages: Identifying Latent Patterns of Behavior

Keith Maki, Michael Miller Yoder, Yohan Jo, Carolyn P. Rosé
Proceedings of the International Joint Conference on Natural Language Processing (IJCNLP), 2017. PDF.

Best Student Paper Award.

Modeling Dialogue Acts with Content Word Filtering and Speaker Preferences

Yohan Jo, Michael Miller Yoder, Hyeju Jang, Carolyn P. Rosé
Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017. PDF.

Metaphor Detection with Topic Transition, Emotion and Cognition in Context

Hyeju Jang, Yohan Jo, Qinlan Shen, Michael Miller, Shane Moon, Carolyn P. Rosé
Proceedings of the Association for Computational Linguistics (ACL), 2016. PDF.

WORKSHOP
PUBLICATIONS

A Fairness Analysis of Human and AI-Generated Student Reflection Summaries

Bhiman Kumar Baghel, Arun Balajiee Lekshmi Narayanan, Michael Miller Yoder
Proceedings of the Workshop on Gender Bias in Natural Language Processing (GeBNLP), 2024. PDF

Identity Construction in a Misogynist Incels Forum

Michael Miller Yoder, Chloe Perry, David West Brown, Kathleen M. Carley, Meredith L. Pruden
Proceedings of the Workshop on Online Abuse and Harms (WOAH), 2023. PDF

Online to Offline Crossover of White Supremacist Propaganda

Ahmad Diab, Bolor-Erdene Jagdagdorj, Lynnette Hui Xian Ng, Yu-Ru Lin, Michael Miller Yoder
International Workshop on Cyber Social Threats (CySoc), Companion Proceedings of the ACM Web Conference (WWW), 2023.

FanfictionNLP: A Text Processing Pipeline for Fanfiction

Michael Miller Yoder, Sopan Khosla, Qinlan Shen, Aakanksha Naik, Huiming Jin, Hariharan Muralidharan, Carolyn P. Rosé
Proceedings of the Workshop on Narrative Understanding (WNU), 2021. PDF.

Code-Switching as a Social Act: The Case of Arabic Wikipedia Talk Pages
Michael Miller Yoder, Shruti Rijhwani, Carolyn P. Rosé, Lori Levin
Proceedings of the Workshop on Natural Language Processing and Computational Social Science (NLP+CSS), 2017. PDF.

WHITEPAPERS &
DEMONSTRATIONS

Misogyny Surfaces as Top Issue in Survey of In-game Experiences
Michael Miller Yoder, Eleanor Young, Jasmin Narisetty, Sam Kulikowski, Will Flanagan, Claire Schuler, Caroline Guo, Zachary Yang
Whitepaper, Collaboratory Against Hate, 2024. PDF.

Research Needs for Countering Extremist Hate
Michael Miller Yoder, Hana Habib
Whitepaper, Collaboratory Against Hate, 2022. PDF.

StoryQ: A Web-Based Machine Learning and Text Mining Tool for K-12 Students
Jie Chao, Bill Finzer, Carolyn P. Rosé, Shiyan Jiang, Michael Miller Yoder, James Fiacco, Chas Murray, Cansu Tatar, Kenia Wiedemann
SIGCSE 2022: Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 2, 1178, 2022.

CONFERENCE
PRESENTATIONS

The ‘unspoken rules’: A multi-platform study of far-right participatory pseudoscientific disinformation
Meredith Pruden, Ayse Lokmanoglu, Michael Miller Yoder
International Communication Association Conference, Cape Town, South Africa, 2026.

Challenges for Real-Time Toxicity Detection in Online Games
Lynnette Hui Xian Ng, Adrian Xuan Wei Lim, Michael Miller Yoder
Ethical Games Conference, virtual, 2024.

The “chanification” of white supremacist extremism
Michael Miller Yoder, Kathleen M. Carley
Conference of the Center for Informed Democracy and Social Cybersecurity (IDeaS), Pittsburgh, PA, 2023.

Digital Chameleons: Identity Signal Changes in COVID-19 Discourse on Twitter
Samantha C. Phillips, Scott Leo Renshaw, Michael Miller Yoder, Kathleen M. Carley
Conference of the Center for Informed Democracy and Social Cybersecurity (IDeaS), Pittsburgh, PA, 2023.

The Relationship Between Antisemitism and COVID-19 Conspiracies on Twitter
Grace Garner, Madeleine McGrann, Daniel Klug, Rachel Kranson, Michael Miller Yoder
International Communication Association Conference, Toronto, Canada, 2023.

The Spatiotemporal Reach of Hate Narratives
Ahmad Diab, Michael Miller Yoder, Yu-Ru Lin
Eradicate Hate Global Summit 2022 Poster Session, Pittsburgh, PA, 2022.

Research Needs for Countering Extremist Hate
Michael Miller Yoder, Hana Habib
Eradicate Hate Global Summit 2022 Poster Session, Pittsburgh, PA, 2022.

**“Everyone is gay and nothing hurts”:
the presentation of queer relationships in fanfiction**

Michael Miller Yoder, Luke Breitfeller, Carolyn P. Rosé
Association for Computers and the Humanities, Pittsburgh, PA, 2019.

Ron the Death Eater: plotting characterization from canon to fanfiction

Michael Miller Yoder, Qinlan Shen, James Fiocco, Carolyn P. Rosé
Society for Text & Discourse, New York, NY, 2019.

Identity and interaction on Tumblr

Michael Miller Yoder, Qinlan Shen, Yansen Wang, Alex Coda, Yunseok Jang, Yale Song, Kapil Thadani, Carolyn P. Rosé
Student Research Symposium
Language Technologies Institute, Carnegie Mellon University, 2019.
Best Poster Award.

Linguistic liminality: Tamazight, Arabic and French in Guigou, Morocco

Michael Miller
Student Research Symposium, Goshen College, 2013.
Audience Choice Award.

INVITED TALKS &
PANELS

**Help, my research area is politicized! Perspectives from
hate speech detection in NLP**

NLP Seminar, University of Pittsburgh, Pittsburgh, PA, 2025.

**Computational approaches to identities and language:
case studies in online extremism**

A.W. Mellon Digital Humanities Workshop, Pittsburgh, PA, 2024.

Using NLP to track white supremacist narratives over time

Carnegie Mellon University Data Science Club, Pittsburgh, PA, 2024.

**Rethinking data: Subjectivity in the data science process and implications
for bias in AI systems**

School of Computing and Information, University of Pittsburgh, Pittsburgh, PA, 2024.

**Algorithms against hate: Mapping online identities and ideologies with
computational techniques**

Department of Mathematics and Computer Science, Duquesne University, Pittsburgh, PA, 2023.

Computational approaches to identities and language

A.W. Mellon Digital Humanities Workshop, Pittsburgh, PA, 2023.

Beyond a hate speech binary

Social Cybersecurity Working Group (virtual), 2022.

**Computational models for social good: Beyond bias and representation
(panel)**

Conference on Social Computing, Behavioral-Cultural Modeling and Prediction and Behavior Representation in Modeling and Simulation (SBP-BRiMS), Pittsburgh, PA, 2022.

Creating a resource directory for crisis (panel)

Eradicate Hate Global Summit, Pittsburgh, PA, 2022.

Detecting hate speech with natural language processing

IdeaS Summer Institute, Carnegie Mellon University, Pittsburgh, PA, 2022.

Computational models of identity presentation in language
Department of Linguistics, University of Pittsburgh, Pittsburgh, PA, 2020.

ADVISING

Thesis committees:

- BPhil Thesis Committee, Michelle Star, “Evaluating Supportive LLM Behavior Across User Contexts Using Simulated Multi-Turn Conversations”, David C. Frederick Honors College, University of Pittsburgh, April 2026
- BPhil Thesis Committee, César Guerra-Solano, “Understanding LLM Performance in Diverse User Contexts”, David C. Frederick Honors College, University of Pittsburgh, March 2026
- Master’s Thesis Committee, Jayden Serenari, “Reducing Sycophancy in Language Models by Leveraging High-Rejection Media and DPO with Concurrent Regularization.”, Department of Computer Science, University of Pittsburgh, December 2025
- Master’s Thesis Committee, Aparna Srinivasan, “Adapting Large Language Models to Genomic Data: Tokenization Methods and Applications for Whole-Genome Sequence Analysis”, Intelligent Systems Program, University of Pittsburgh, April 2025
- Master’s Thesis Committee, Deyasini Mitra, “Transfer Learning in Molecular Data Analysis: Unleashing the Potential of Large-Language Models”, Intelligent Systems Program, University of Pittsburgh, April 2025
- Master’s Thesis Committee, Raja Krishnaswamy, “A Framework for Concept Extraction from Course Presentations”, Department of Computer Science, University of Pittsburgh, April 2024

Capstone projects:

- Undergraduate capstone project, Titus Kurian, “Concept Extraction Dataset from Presentation Slides of Computer Science Courses”, Department of Computer Science, University of Pittsburgh, April 2026
- Master’s capstone project, John Bowen, “Enhancing Document-Level English Machine Translation for Korean Literature”, Department of Computer Science, University of Pittsburgh, April 2025

HONORS &
AWARDS

Best Reviewer Award
EMNLP 2023 Computational Social Science and Cultural Analytics

Outstanding Reviewer Award
ACL 2021

Best Poster Award
2019
Student Research Symposium
Language Technologies Institute, Carnegie Mellon University

Best Student Paper Award
2017
International Joint Conference on Natural Language Processing (IJCNLP)

NSF Graduate Research Fellowship
2015–2018
Awarded national fellowship with a focus area in natural language processing.

Audience Choice Award
2013
Student Research Symposium, Goshen College

President's Leadership Award Scholarship, Goshen College
2009–2013

Awarded top merit scholarship for leadership in academic studies.

ACADEMIC
SERVICE

Organizing:

- Collaboratory Against Hate Workshop on Practitioner Research Needs, 2022
- Eradicate Hate Global Summit 2022 Poster Session

Committees:

- Advisor, Technical Track, Hacking 4 Humanity 2025
- Judge Panel, Hacking 4 Humanity 2023
- Co-Director Search Committee, Collaboratory Against Hate, 2023
- Diversity, Equity, and Inclusion Committee, CMU Language Technologies Institute, 2020-2021
- Anti-Racism Working Group, CMU School of Computer Science Dean's PhD Advisory Committee, 2020-2021
- Update.org Data Science for Social Good Fellows Mentor Committee, 2019

Reviewing (NLP and Computational Social Science):

- ACL Rolling Review (2022-)
- CHI 2025
- SPELL 2023
- *ACM Journal on Computing & Sustainable Societies*, 2023
- *EPJ Data Science*, 2023
- *Computational and Mathematical Organization Theory*, 2022, 2024
- USEC 2022
- EMNLP Computational Social Science and Cultural Analytics track 2022, 2023 (**Best Reviewer Award**)
- ACL 2021 (**Outstanding Reviewer Award**), 2023
- CoNLL 2020, 2022 Understanding LLM Performance in Diverse User Contexts
- Bridging Human-Computer Interaction and NLP Workshop (HCI+NLP) 2021
- NAACL, AACL and ACL Student Research Workshops 2020, 2021, 2022, 2024
- Human and Machine-in-the-Loop Evaluation and Learning Strategies Workshop (HAMLETS @ NeurIPS) 2020

Reviewing (Digital Humanities):

- *Between Invention and Audience: DocuScope and the Rhetorical Approach to Corpus Analysis* (Studies in Corpus Linguistics, John Benjamins Press), 2022
- *Humanities* journal, special issue on fanfiction, 2022
- Association for Computers and the Humanities (ACH) 2021
- Digital Humanities (DH) 2020

INTERNATIONAL
EXPERIENCE

Bicycle Tourist

2013–2014

Paraguay, South America to Indiana, USA

Bicycled with a group of three over 10,000 miles from South America to the United States on an 11-month self-supported journey. Learned conversational Spanish and gained familiarity with Latin American culture.

Researcher and Service Worker

2012

Meknes and Guigou, Morocco

Studied Moroccan language and culture for 12 weeks while living with host families in urban and rural settings. Researched code-switching among Tamazight, French, Colloquial and Modern Standard Arabic speakers in the Middle Atlas Mountains.

Youth Camp Volunteer

2011

West Bank, Palestine & Israel

Studied and practiced Levantine and Modern Standard Arabic for 9 weeks. Participated in history and learning tours of the Israeli-Palestinian conflict while living with a Palestinian friend's family. Facilitated crafts and games for Palestinian youth.